

Chocolate-brown blood + saturation gap after dapsone, benzocaine, or nitrites

Dx: Methemoglobinemia: chocolate-brown blood plus saturation gap after dapsone, benzocaine, or nitrites

RULE: Pulse oximetry may stay near 85% despite normal/high PaO₂

TREAT: Methylene blue; avoid in G6PD deficiency and use high-dose ascorbic acid or exchange transfusion instead

Tinnitus + tachypnea + respiratory alkalosis and anion-gap metabolic acidosis

Dx: Salicylate toxicity: tinnitus plus respiratory alkalosis and anion-gap metabolic acidosis

TREAT: IV sodium bicarbonate for urinary alkalinization; hemodialysis for level >90 mg/dL, severe neurologic symptoms, pulmonary edema, or renal failure

TRAP: Intubation can be fatal if minute ventilation does not match the patient's respiratory drive

Long-bone fracture + hypoxemia, confusion, and petechial rash within 24-72 hours

Dx: Fat embolism syndrome: long-bone fracture plus hypoxemia, confusion, and petechial rash within 24-72 hours

TREAT: Early fracture stabilization and supportive care

CONTRAST: Post-op PE causes pleuritic pain/tachycardia but not the classic petechial rash/timing triad

Nephrology

Urinalysis, Hematuria, Proteinuria, and Biopsy

Dysmorphic RBCs, RBC casts, brown/cola urine

Dx: Glomerular hematuria / glomerulonephritis pattern

NEXT: Nephrology workup; kidney biopsy is indicated if there is active sediment, declining renal function, or systemic disease

CONTRAST: Bright red blood/clots with **isomorphic RBCs** suggests a urologic source

≥3 RBCs/hpf on urine microscopy

RULE: True microscopic hematuria is **≥3 RBCs/hpf** on microscopy from a properly collected, non-acute specimen

NEXT: Confirm microscopy, exclude benign causes, then risk-stratify; higher-risk patients need cystoscopy and upper-tract imaging

CONTRAST: Proteinuria, dysmorphic RBCs, or RBC casts points to glomerular disease and needs nephrology evaluation; anticoagulation does not remove the need to evaluate hematuria

Hematuria with clots

Dx: Lower urinary tract bleeding is more likely, especially bladder or collecting-system source

CONTRAST: Glomerular hematuria usually has dysmorphic RBCs/RBC casts and typically does **not** form clots